

ROS 2 Tutorial

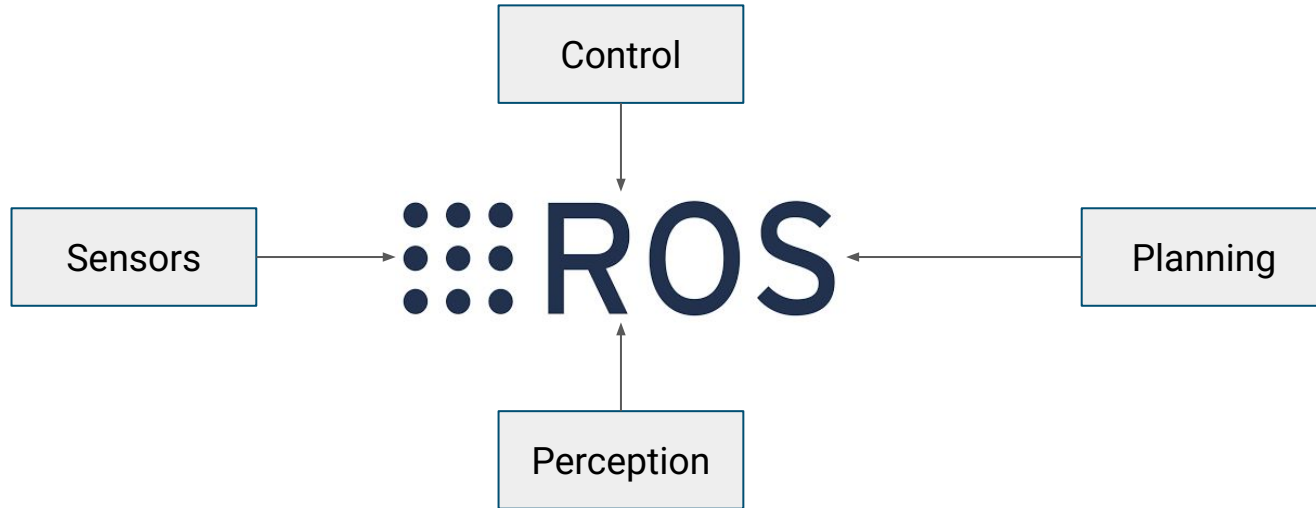
Jing-Yan Liao, Chenghao Li

Outline

- What is ROS? Why ROS?
- Core Concepts of ROS 2
- Useful Tools & Packages
- Useful references

What is ROS?

- Robot Operating System, but not your typical OS
- Middleware that glues all parts of robots together



ROS vs ROS 2



- Beginning of ROS (EOL 2025)



- Support Real-time
- Multi-robot communication
- Cross-platform (Linux, Windows, macOS, embedded)
- Better security

Why ROS?

- Standardize communication between modules
- Huge collections of packages (drivers, SLAM, navigation, manipulation,...)
- If you want to get a job...

Tldr; No reinventing the wheel for every robot + Every robotics company uses ROS

Core concepts in ROS/ROS 2

- Node
 - A process that perform a specific task
 - Ex: LiDAR driver node, object detection node, or motor controller node
 - Robots will run several nodes at once

Core concepts in ROS/ROS 2

- Node
 - A process that perform a specific task
- Topic
 - A channel for **streaming data** between nodes
 - Publisher
 - Subscribers
 - Example:
 - LiDAR node publish point cloud on `/lidar/points_raw`
 - Object detection subscribe to `/lidar/points_raw`

Core concepts in ROS/ROS 2

- Node
 - A process that perform a specific task
- Topic
 - A channel for **streaming data** between nodes
- Message
 - Data Structure exchange over topics
 - Make sure ros2 domain id is the same

sensor_msgs/PointCloud2 Message

File: `sensor_msgs/PointCloud2.msg`

Raw Message Definition

```
# This message holds a collection of N-dimensional points, which may
# contain additional information such as normals, intensity, etc. The
# point data is stored as a binary blob, its layout described by the
# contents of the "fields" array.

# The point cloud data may be organized 2d (image-like) or 1d
# (unordered). Point clouds organized as 2d images may be produced by
# camera depth sensors such as stereo or time-of-flight.

# Time of sensor data acquisition, and the coordinate frame ID (for 3d
# points).
Header header

# 2D structure of the point cloud. If the cloud is unordered, height is
# 1 and width is the length of the point cloud.
uint32 height
uint32 width

# Describes the channels and their layout in the binary data blob.
PointField[] fields

bool    is_bigendian # Is this data bigendian?
uint32  point_step   # Length of a point in bytes
uint32  row_step      # Length of a row in bytes
uint8[] data          # Actual point data, size is (row_step*height)

bool is_dense         # True if there are no invalid points
```

Ref: ROS documentation

Core concepts in ROS/ROS 2

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- Service
 - A channel only provide data upon request
 - Example:
 - Request: SaveMap
 - Response: MapSavedSuccess (bool)

Core concepts in ROS/ROS 2

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- Message
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- Service
 - A channel only provide data upon request
- Launch file
 - A script to start multiple nodes with configured parameters all at once

Core concepts in ROS/ROS 2

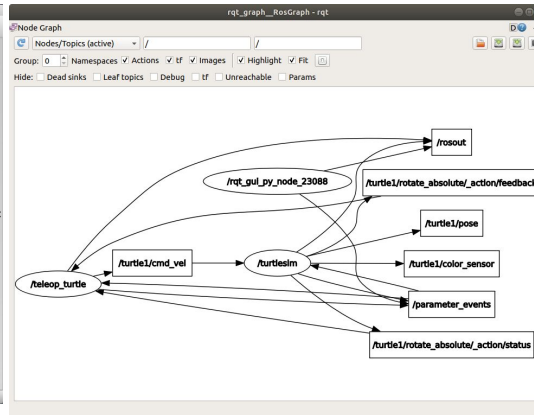
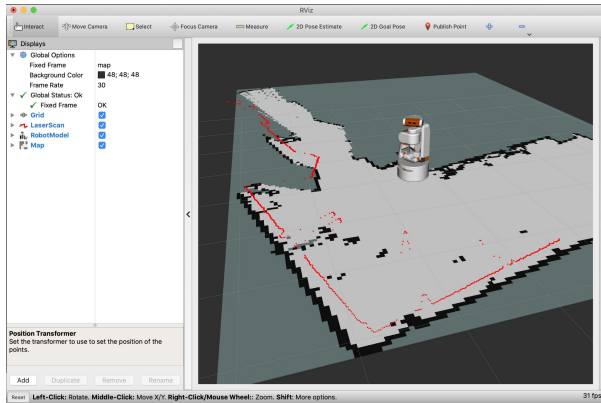
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Useful tools / packages

- **tmux**
 - A tool that can manage multiple terminal sessions in a single window, even after disconnecting
- **Rviz2 / foxglove**
 - ros2 visualization tools that can
 - visualize sensor data
 - overlay output on top of sensor data

Visualization Tools

- RVIZ2 – 3D visualization of robot data
- RQT (ROS2 QT) – Flexible GUI for debugging and monitoring
- Foxglove – Modern web-based visualization and analysis



RVIZ2

What data does RVIZ2 visualize?

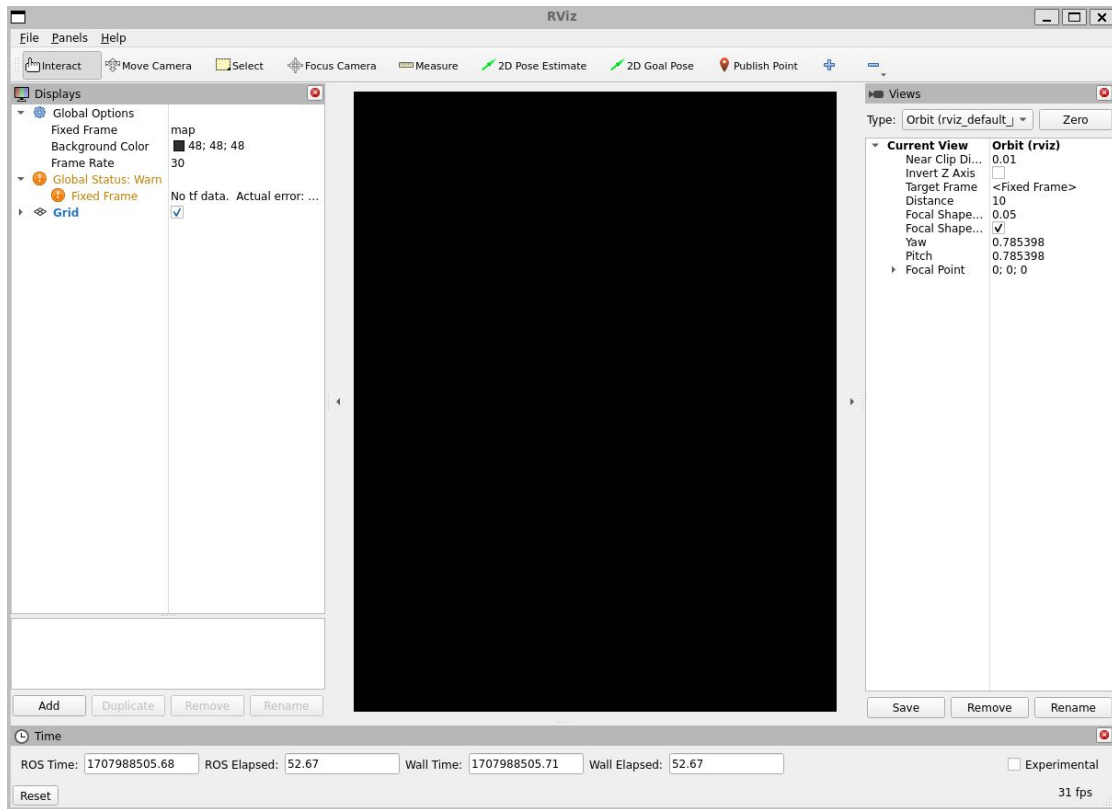
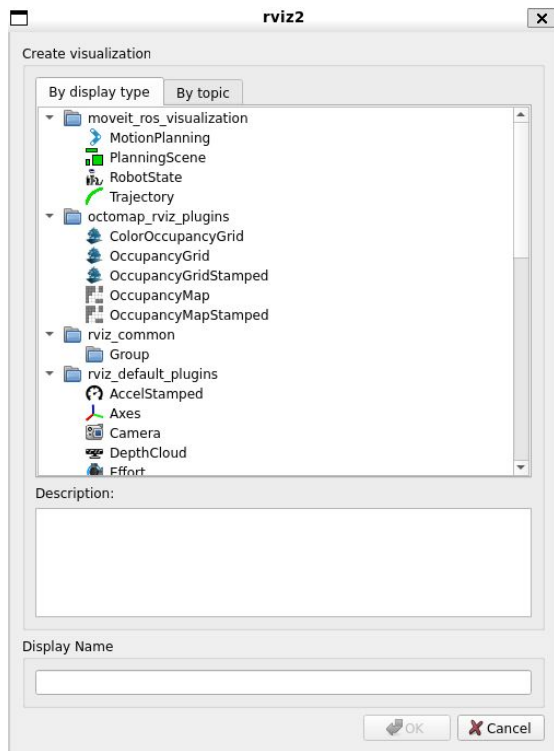
Image, Point Cloud, Map, TF(transform), Robot model。

Why we need visualization?

Making abstract data intuitive and visible.



Plugin-based design.

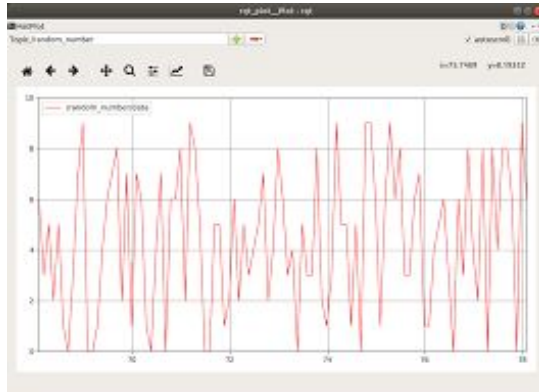


RQT

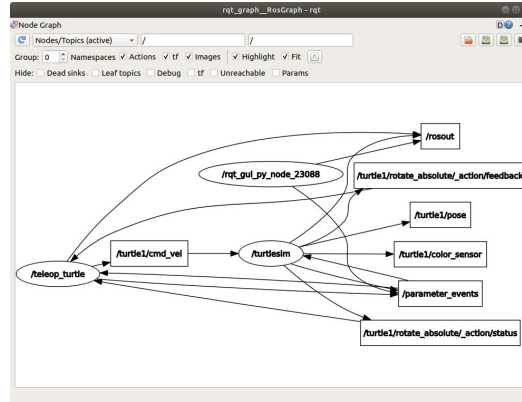
RQt is an another graphical user interface (GUI) framework in ROS2.

RVIZ2 -> focuses on 3D data visualization (maps, point clouds, robot models)

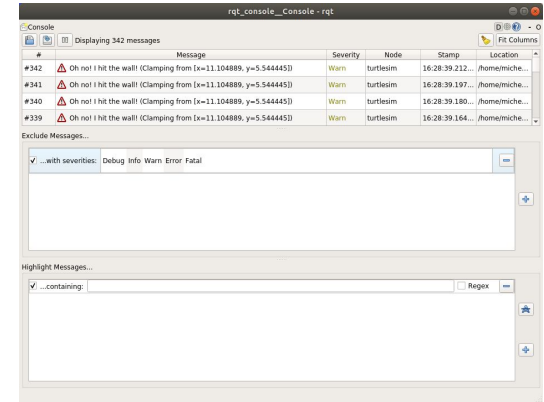
RQT -> focuses on GUI tools for debugging & monitoring (plots, topic inspection, logs)



plots



nodes

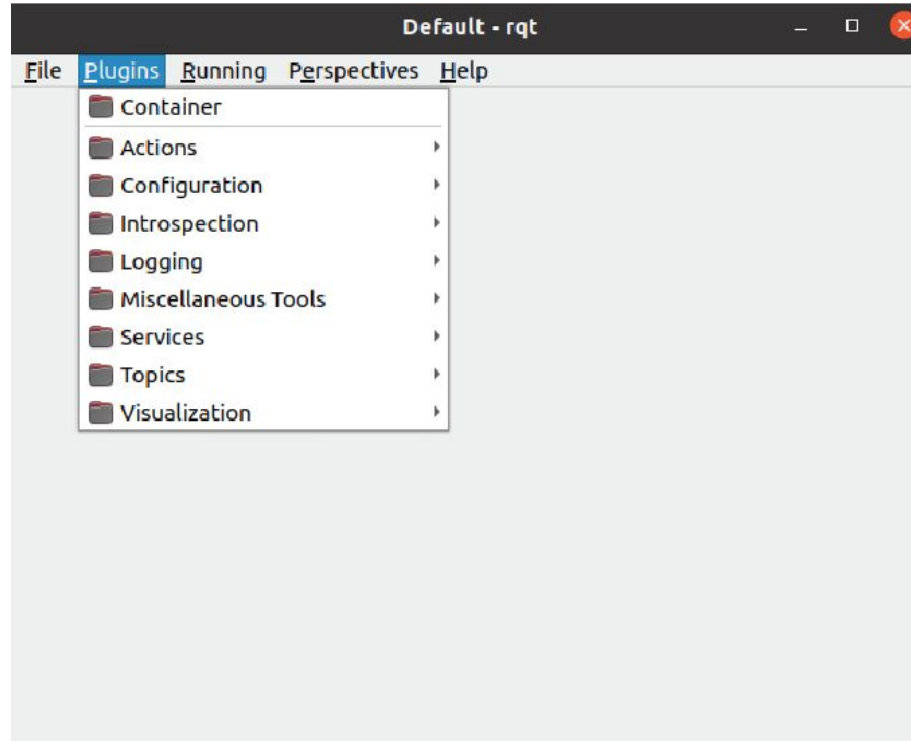
A screenshot of the RQT Console window. The title bar reads 'rqt_console - rqt'. It displays a table of log messages. The table has columns for #, Message, Severity, Node, Stamp, and Location. The messages are warnings from the 'turtle1' node about clamping coordinates.

#	Message	Severity	Node	Stamp	Location
#342	Oh no! I hit the wall! (Clamping from [x=11.104889, y=5.544445])	Warn	turtle1sim	16:28:39.212...	/home/miche...
#341	Oh no! I hit the wall! (Clamping from [x=11.104889, y=5.544445])	Warn	turtle1sim	16:28:39.197...	/home/miche...
#340	Oh no! I hit the wall! (Clamping from [x=11.104889, y=5.544445])	Warn	turtle1sim	16:28:39.180...	/home/miche...
#339	Oh no! I hit the wall! (Clamping from [x=11.104889, y=5.544445])	Warn	turtle1sim	16:28:39.164...	/home/miche...

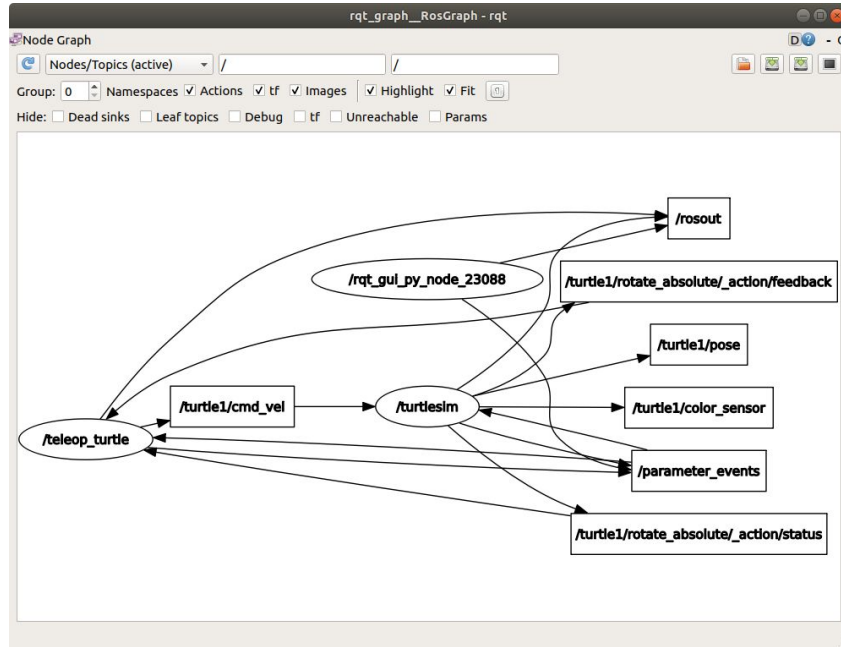
logs

RQT

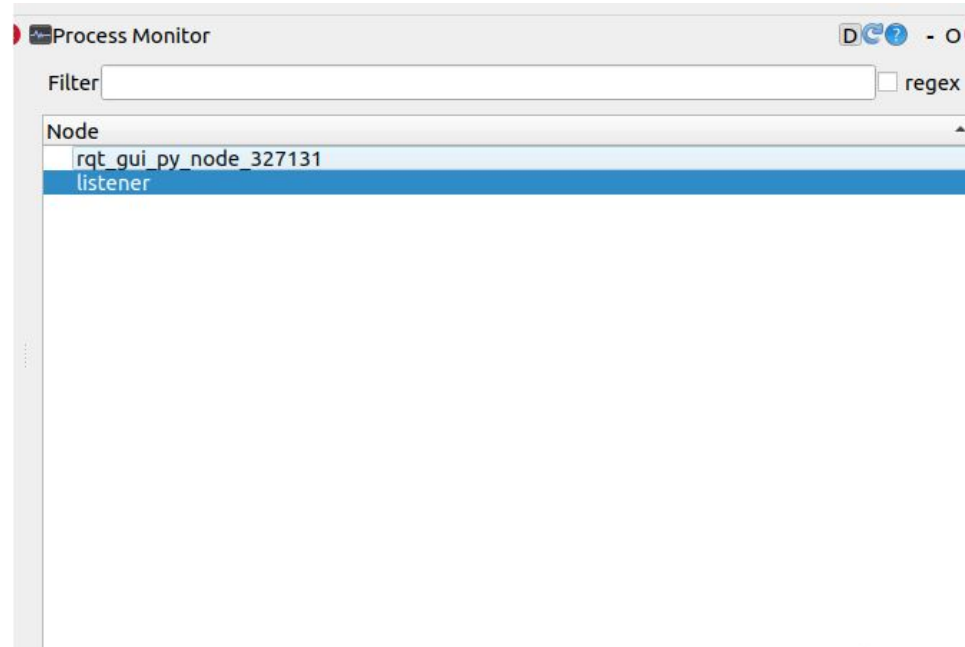
Also plugin-based design.



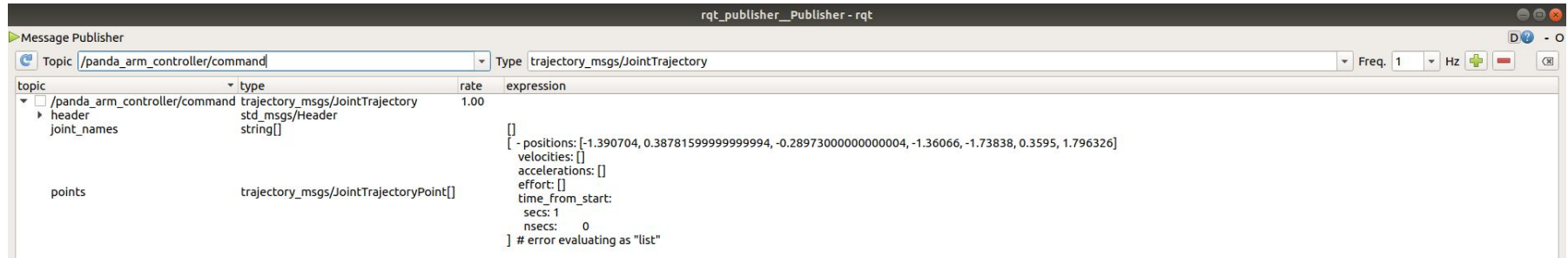
Introspection / Node Graph



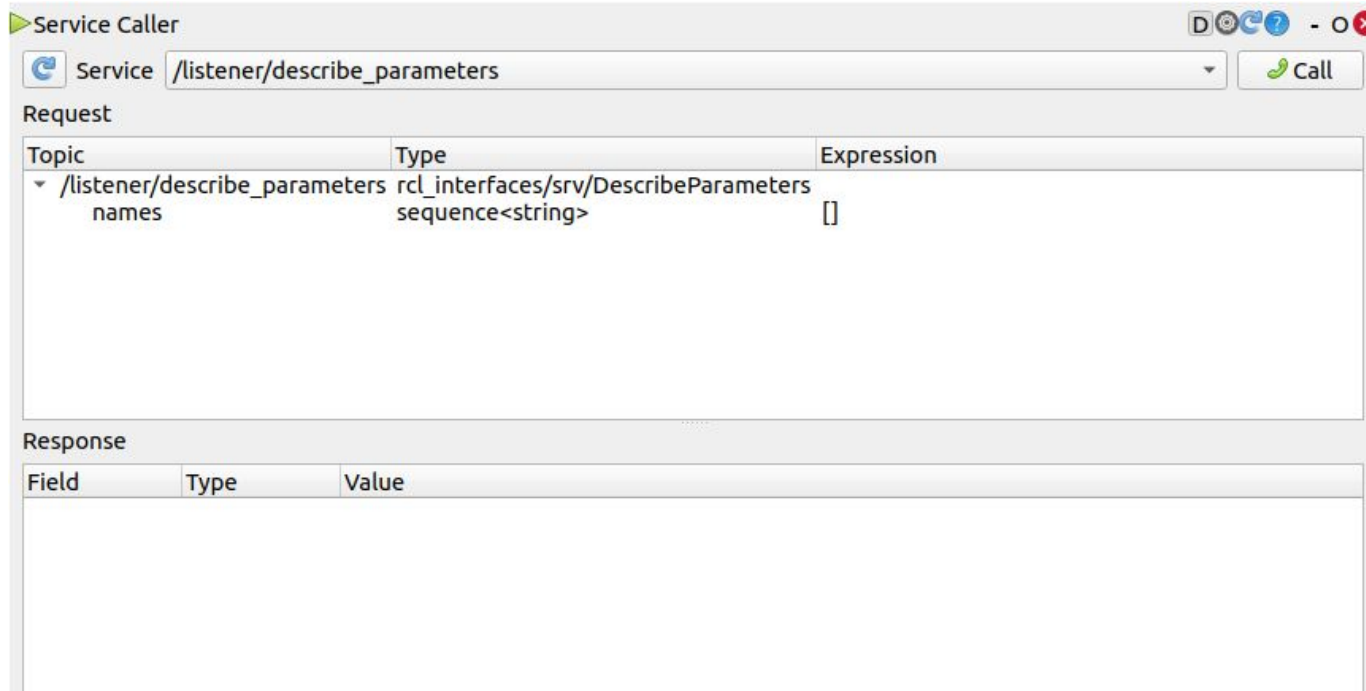
Introspection / Process Monitor



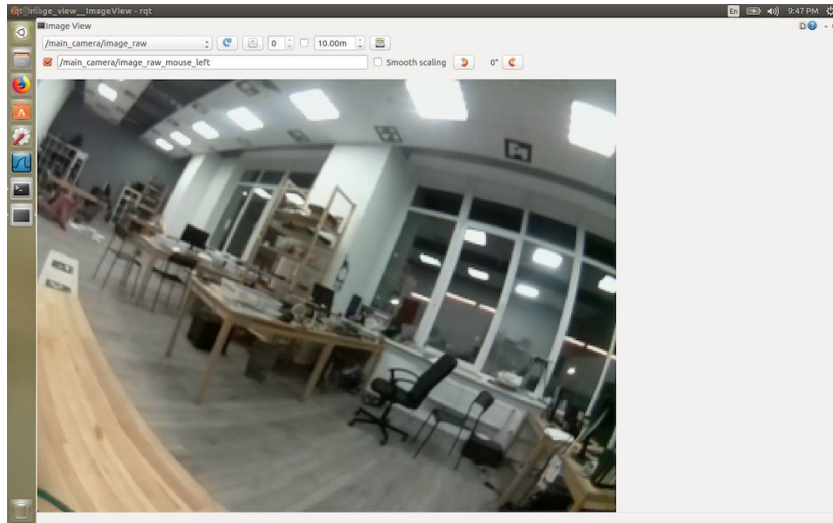
Topic/ Message Publisher



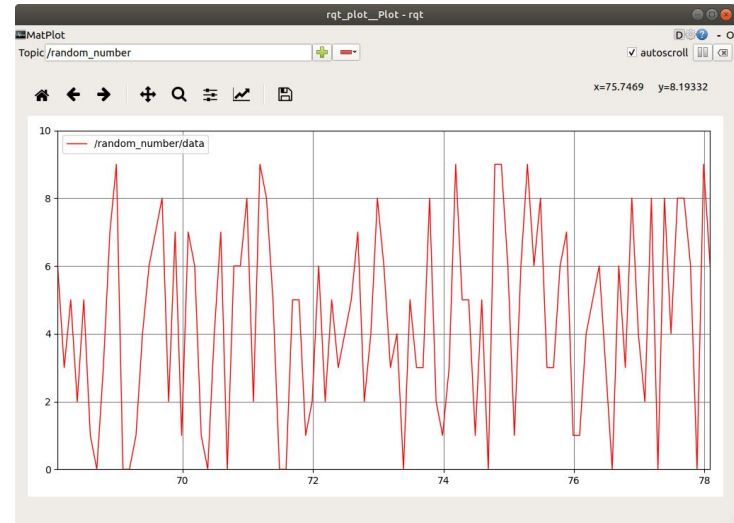
Service /Service Caller



Image



Matplot



Foxglove - A 3rd-party tool

Web-based & cross-platform.

Multi-system support.

Data logging & playback.

Open source.

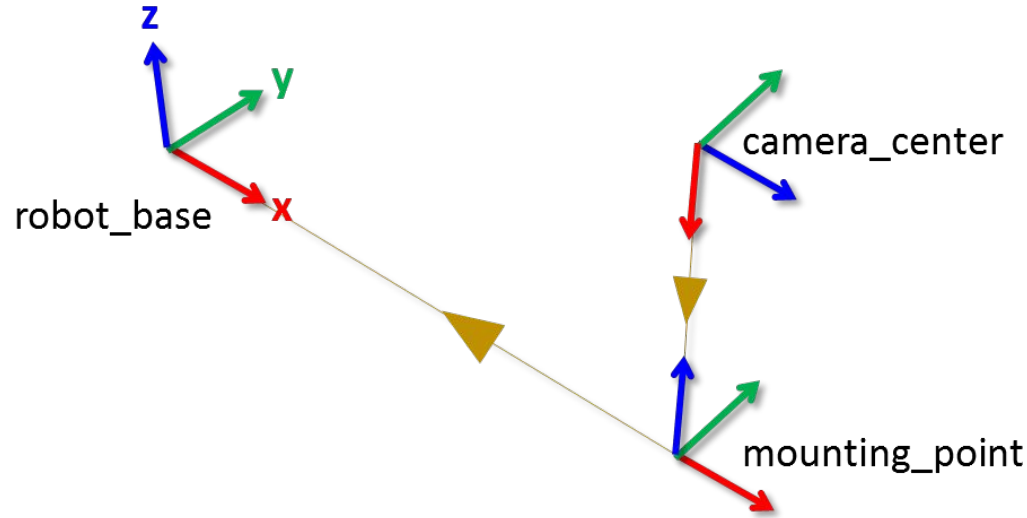


Frame

A frame = a coordinate system

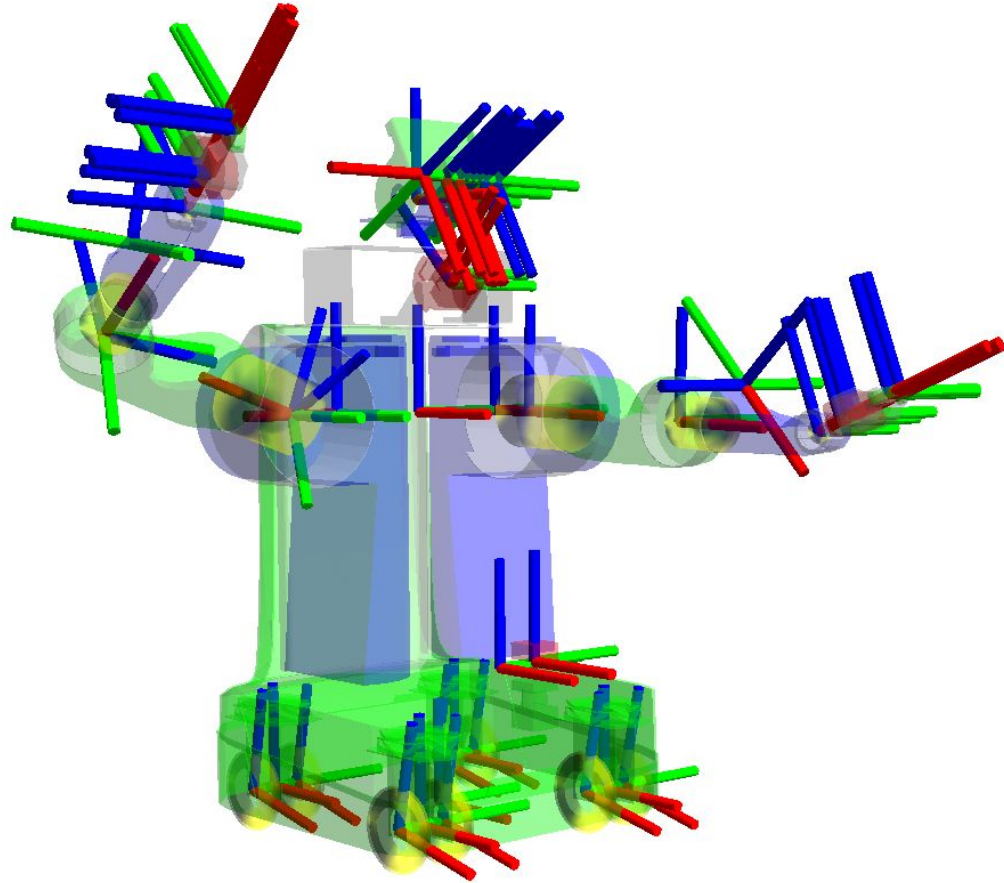
Includes position and orientation information.

Use transformation matrices (rotation + translation) to convert between frames



TF2 (TransForm)

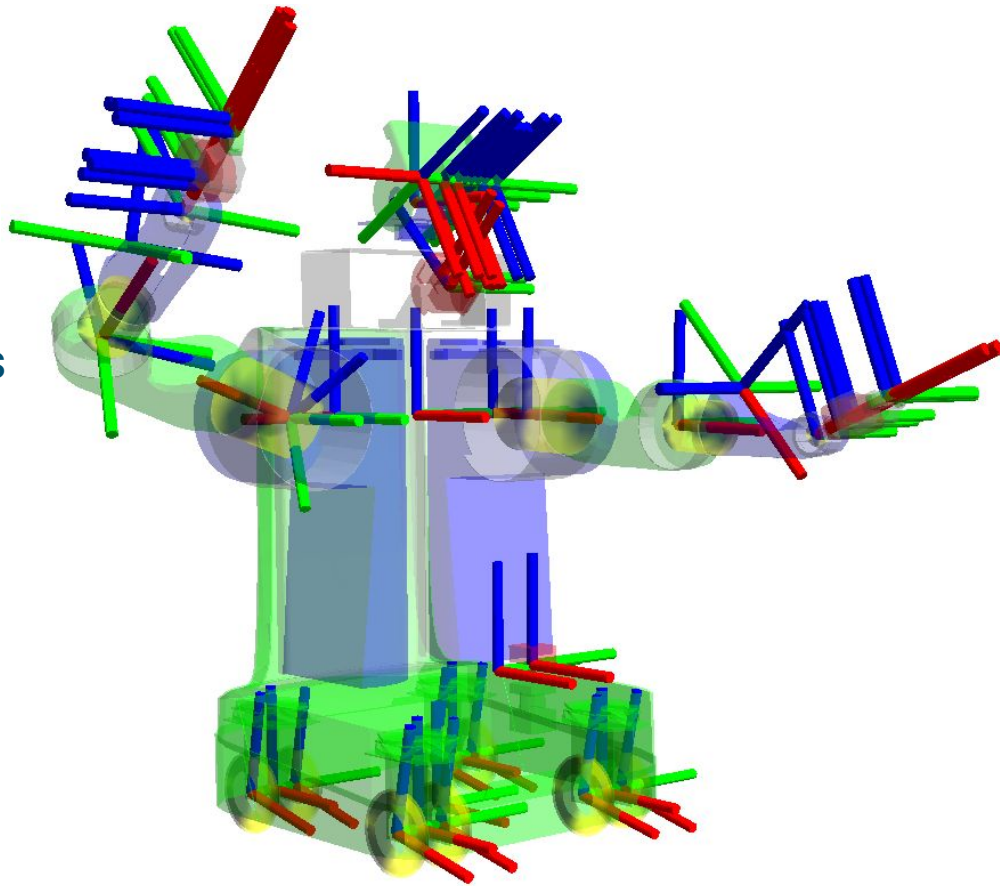
Lets try this.



TF2 (TransForm)

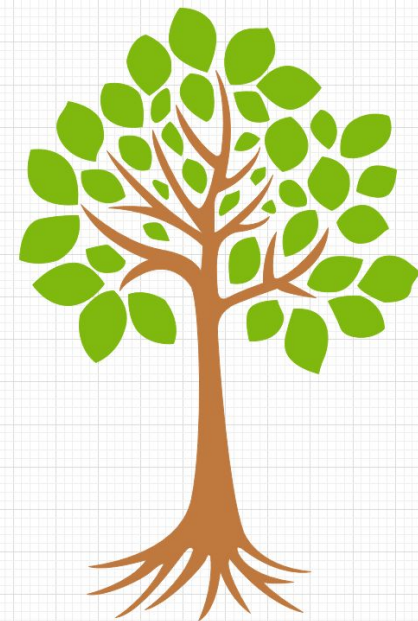
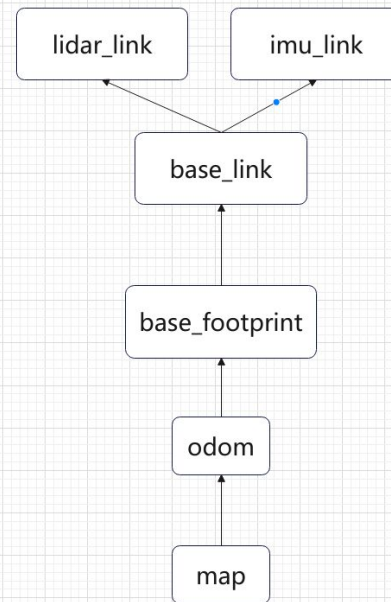
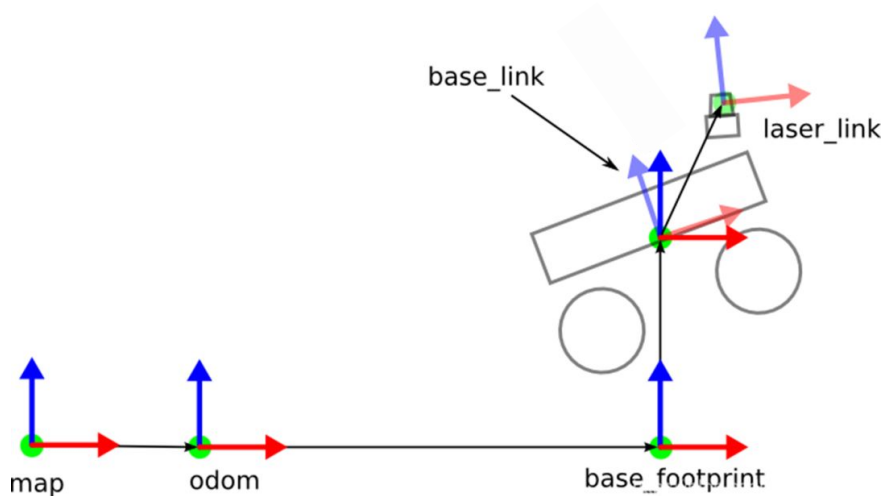
Lets try this.

TF2 is the transform library, which lets the user keep track of multiple coordinate frames over time.



TF2 (TransForm)

TF Tree Example



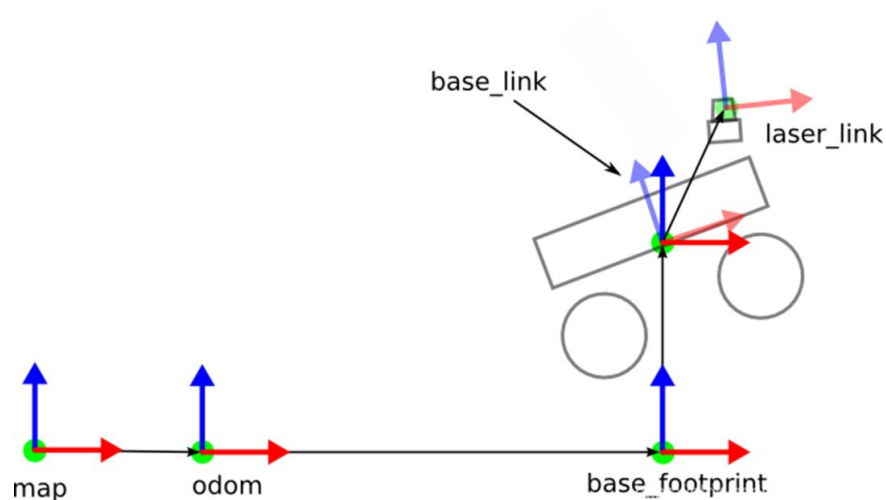
TF2 (TransForm)

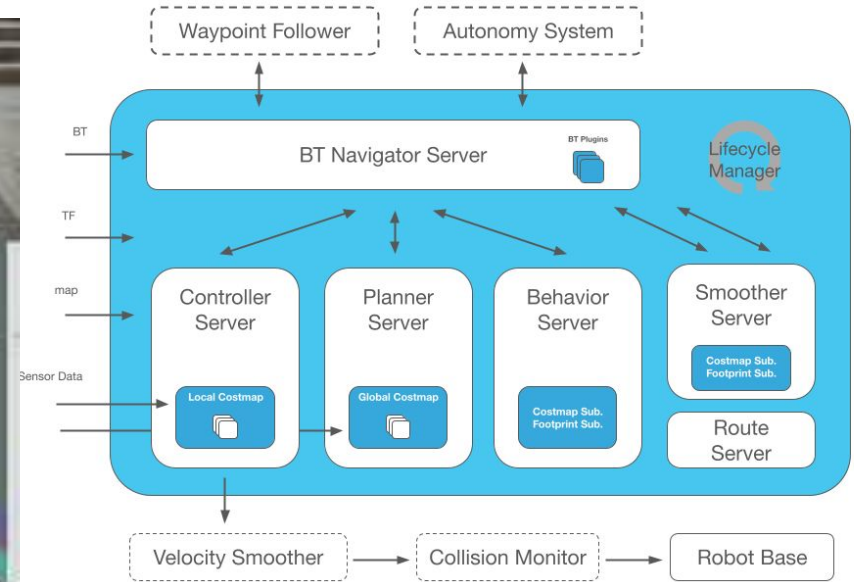
Static TF

By defining transformations between the base_link frame (robot frame) and each sensor frame

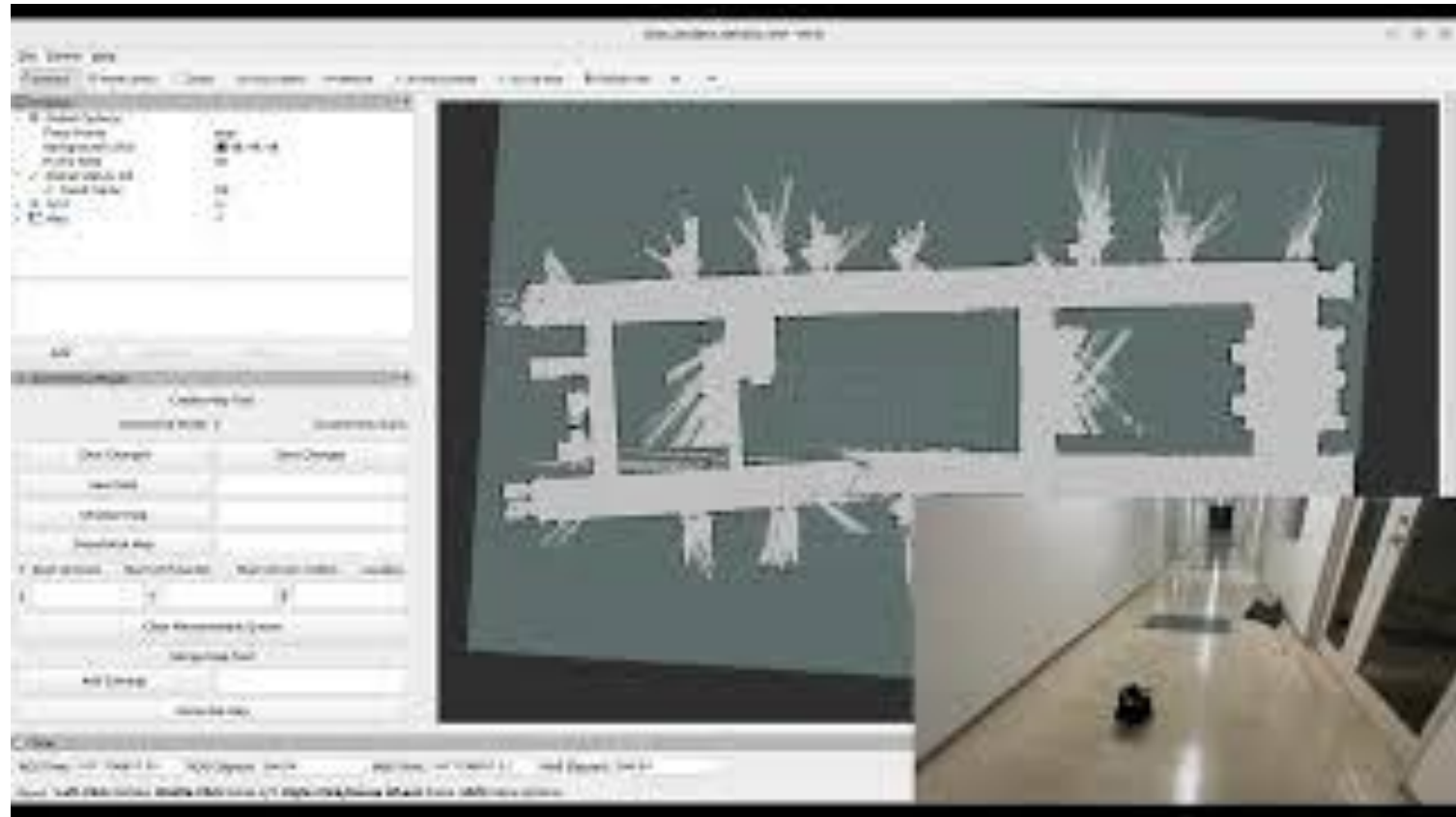
Dynamic TF

Used for moving joints, wheels, manipulators

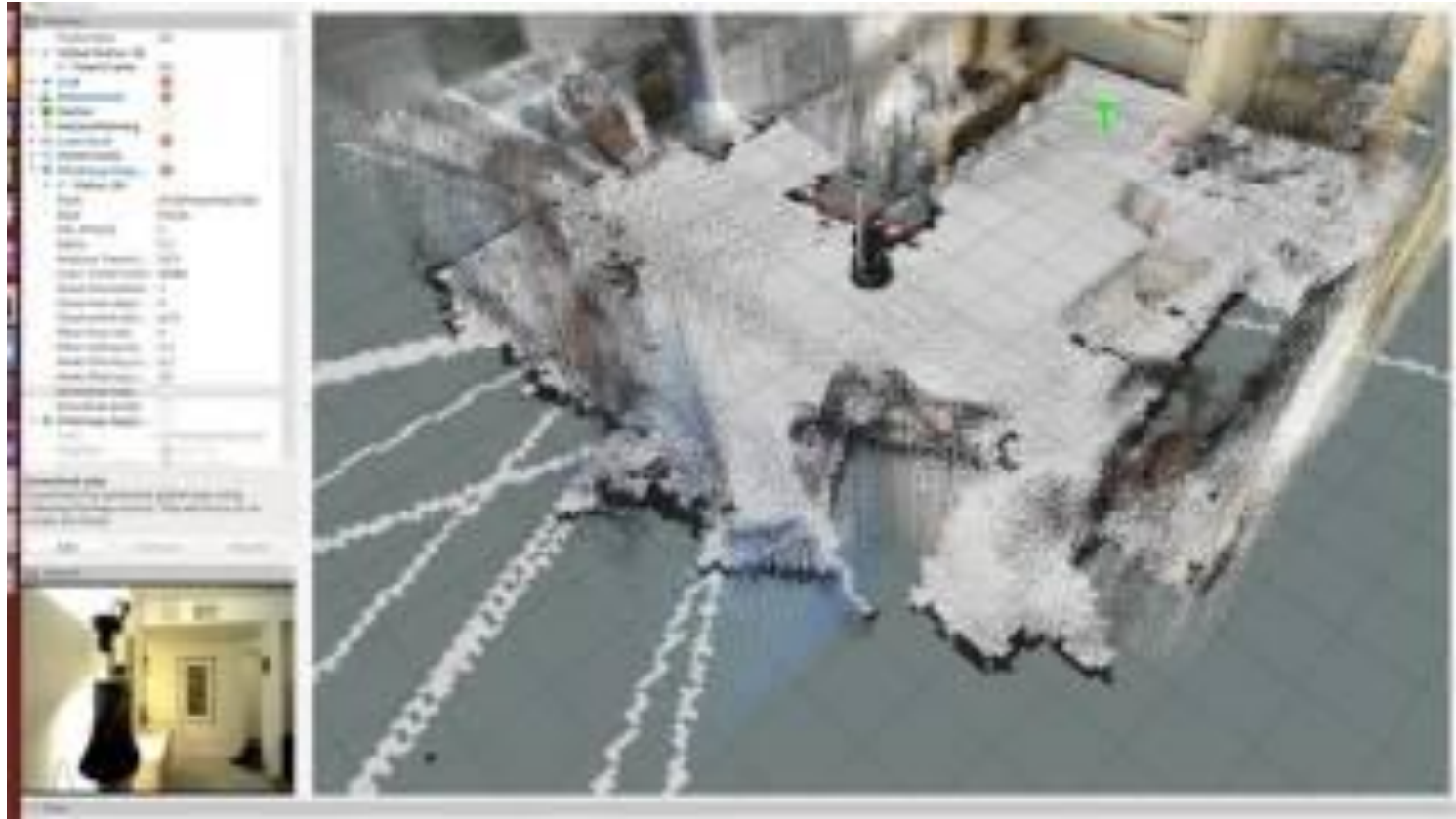




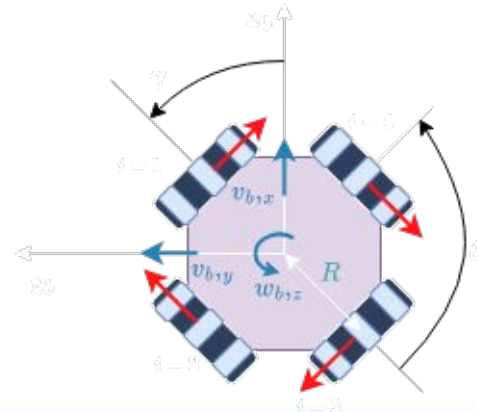
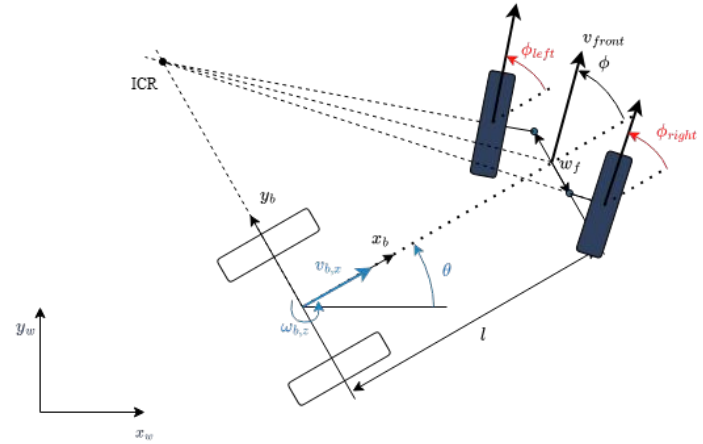
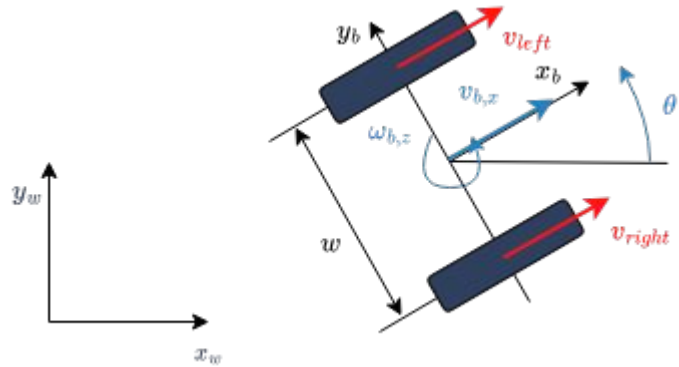
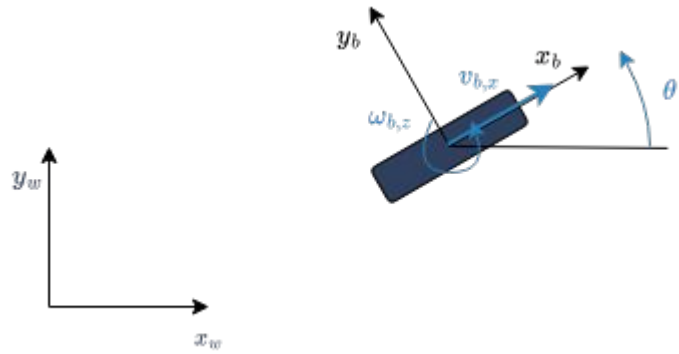
slam_toolbox



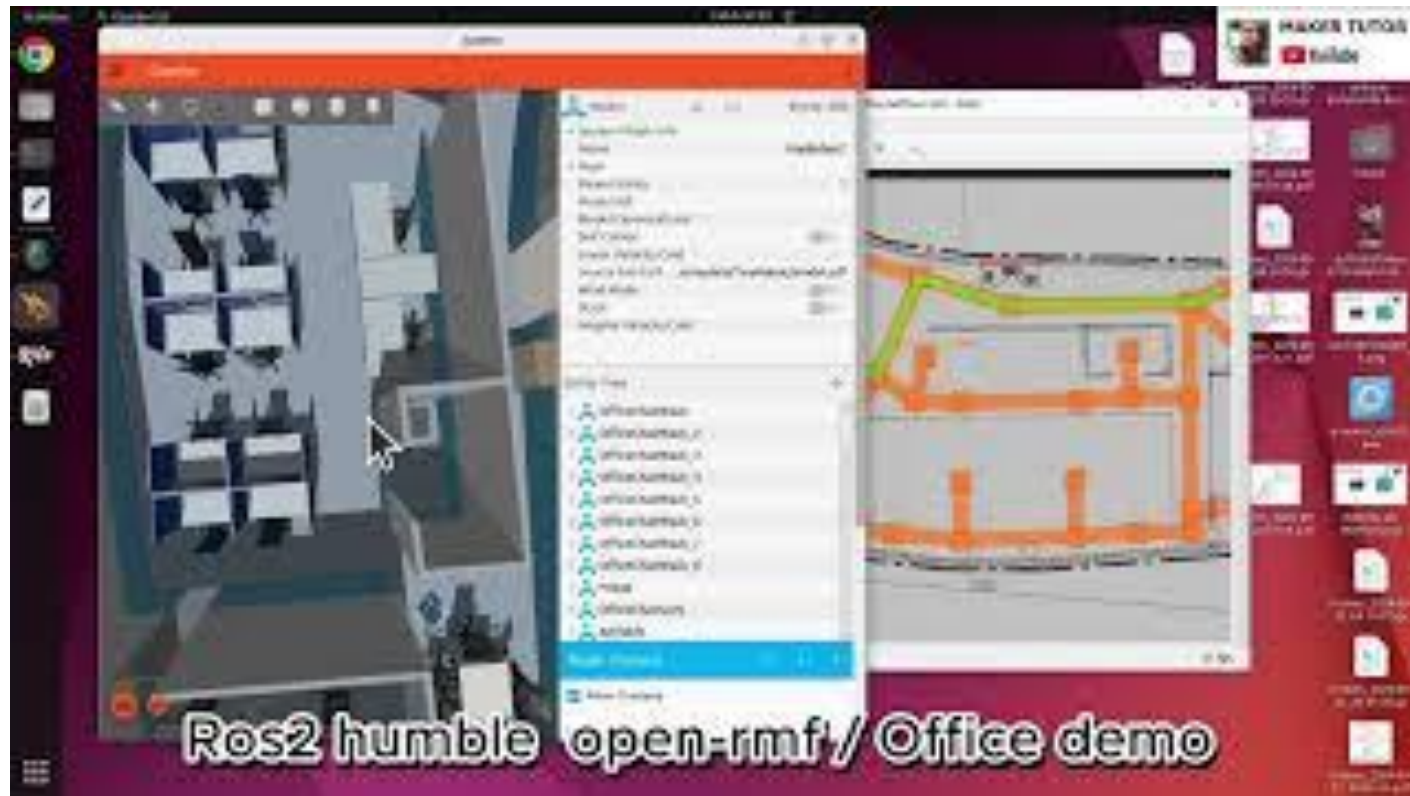
RTAB-Map



ros2_control



Open-RMF



Virtual Machine

Windows -> Dual boot systems

-> virtualbox

MacOS -> virtualbox or UTM

The Power of the ROS2 Community

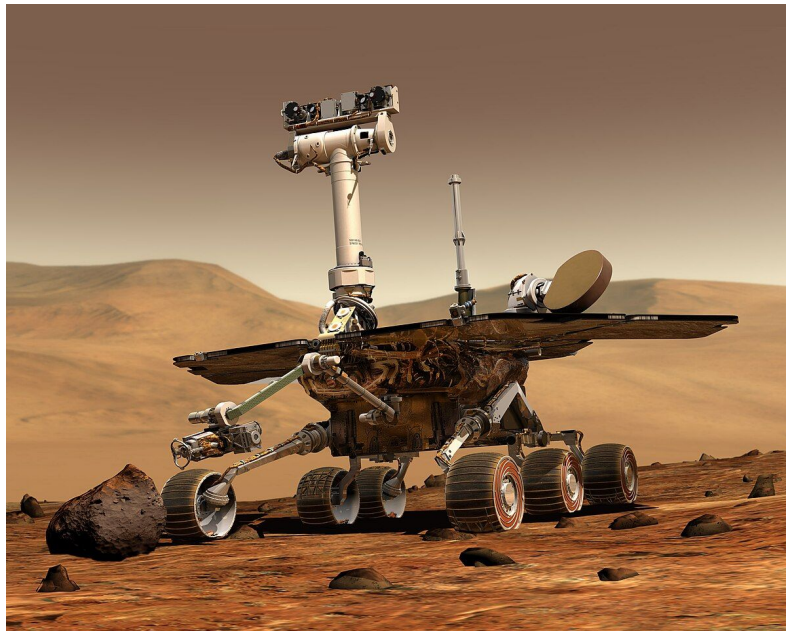
- Thousands of developers worldwide
- Growing ecosystem of packages and tools
- Continuous improvements and new features
- Open source: anyone can contribute
- You can be part of it too! 🚀



Congratulations, you now know ROS2!

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HW1: build a Mars Rover with ROS2...



Demo Time

Tmux -> [link](#)

Turtlesim -> [link](#)

TF2 -> [link](#)

Nav2 -> [link](#)

Reference

Waverover robot: [link](#)

Rubik Pi 3: [link](#)

ROS2 Jazzy: [link](#)

Nav2: [link](#)

Slam_toolbox: [link](#)

RTABMap: [link](#)

Open-RMF: [link](#)

Virtual Box: [link](#)

Dual Boot: [link](#)

UTM: [link](#) UTM Guide: [link](#)

[Tmux cheat sheet](#)